

Device: iPhone 15 Pro

Our response to the quality requirements you provided:

- **0.5x–0.6x zoom (wide)**

→ Does not meet in the iPhone RGB+LiDAR+IMU configuration. iOS does not expose the 0.5x camera as the RGB stream in the same ARKit session. Meeting the 0.5x requirement requires a separate RGB+IMU configuration using the iPhone 15 Pro Ultra Wide camera, without LiDAR depth or ARKit camera pose.

- **Landscape orientation, 1080p, 30fps, 8-bit (HDR off)**

→ Meets. The delivered footage is landscape 1920×1080, 30fps, H.264 8-bit SDR, with HDR disabled.

- **Head-mounted, at forehead/eye level, angled slightly down**

→ Meets. The iPhone is secured in a head mount at forehead/eye level with the camera angled slightly downward toward the hands and work area.

- **Hands visible in at least 80% of frames — we measure this per clip and it is the most common reason footage comes back**

→ Meets through per-clip QC. Hand visibility is measured for every clip, and clips below the 80% threshold are excluded from delivery.

- **RGB and IMU on a single hardware clock source, with the per-frame video↔IMU timestamps delivered alongside each clip**

→ Meets. Video frames use ARFrame capture timestamps and IMU samples use Core Motion timestamps. Both are expressed on the same monotonic device-uptime timeline. Each video frame is delivered with its timestamp in frames.jsonl, and the raw accelerometer and gyroscope samples are delivered with their timestamps in imu.jsonl.

Related, and useful for us to know early: what video-to-IMU time offset does your stack actually measure, and how do you record it per clip? We would rather work to a figure you can evidence than pick one in the abstract.

→ Our capture stack records the acquisition timestamps supplied by ARKit and Core Motion directly, without applying an additional fixed offset. On-device validation of this configuration on the iPhone 15 Pro, using temporal cross-correlation between inter-frame image motion and gyroscope magnitude, measured a video-to-IMU residual of approximately −2.93 ms, using the convention $t_{\text{IMU}} - t_{\text{video}}$. The measured video/gyroscope correlation was 0.96, with a 3.40 ms standard deviation across independent time windows. Because the absolute residual is smaller than the observed window-to-window variation, applying a fixed timestamp shift is not expected to improve alignment. The original acquisition timestamps are therefore delivered unchanged: frames.jsonl records the timestamp of every delivered video frame, and imu.jsonl records the raw accelerometer and gyroscope timestamps on the same timeline. These figures apply to the RGB+LiDAR+IMU configuration used for the sample; a separate 0.5x RGB+IMU configuration would require its own validation.

On the iPhone 15 Pro the LiDAR is aligned to the wide camera, while 0.5x is the ultra-wide. How do you handle depth coverage at 0.5x — is the depth reprojected into the ultra-wide frame, or does that combination force a trade-off between the wider field of view and full-frame depth?

→ iOS does not support recording the 0.5x camera and ARKit depth in the same session. It also does not provide a supported way to run a separate 0.5x camera session in parallel with an ARKit session. The depth is therefore not reprojected into a 0.5x frame. If both 0.5x–0.6x RGB and LiDAR depth are required, the iPhone 15 Pro cannot meet the combined requirement. Meeting the wide-angle requirement requires a separate 0.5x RGB+IMU configuration without LiDAR depth or camera pose.